1. Retargeting Social Media Outreach for SNAP Take-Up

We use the public deidentified replication data released with Rogers (2026) in Management Science. The outcome is SNAP enrollment in the March 1, 2022 snapshot. We select one English-language retargeting advertisement (raw group 2) and its English-language control (raw group 4), leaving 7,110 individuals.

Groups use baseline SNAP enrollment, household-size measures and baseline benefit measures. Median splits with randomized ties and a threshold of more than 100 observations yield $K = 31$ groups containing 4,511 records from 7,110 complete input records.

The mean of the group benchmark HTEs is $\tau_0 = 0.007493$. Their distribution is shown in Figure 1.

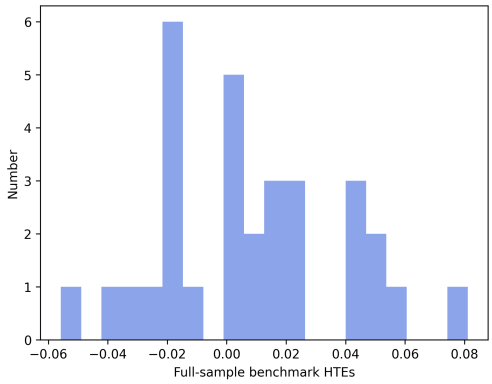
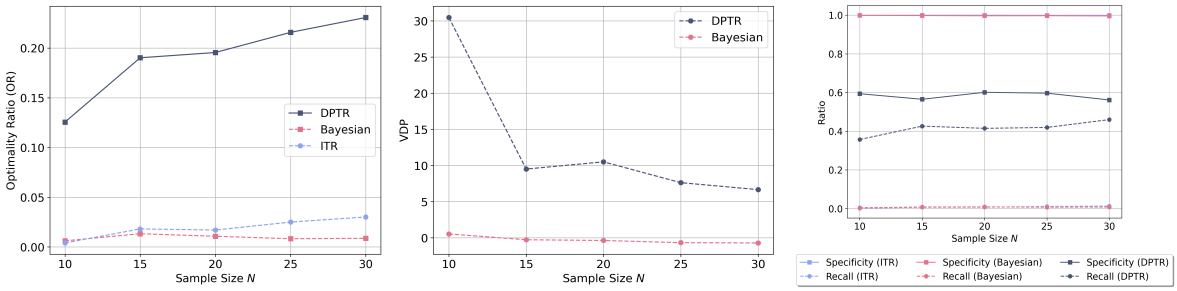


Figure 1 **Histogram of full-sample benchmark HTEs at Rogers (2026).**

Figure 2 reports mean performance under the common resampling protocol described above.



(a) OR (b) VDP (c) Recall and Specificity

Figure 2 **Performance comparisons at Rogers (2026) under different sample sizes N.**

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

The ITR and Bayesian recall curves nearly overlap at the plotted scale, as do their specificity curves. Assignment occurred at household level, whereas the bootstrap resamples individuals.

2. Using Social Recognition to Address the Gender Difference in Volunteering for Low-Promotability Tasks

We use the public laboratory replication data released with Banerjee and Mustafi (2026) in Management Science. We select the Baseline and Positive recognition arms and the ten gameplay rounds, giving 3,000 participant-round records. Treatment announces and applauds the volunteer's real name. The outcome indicates an individual volunteering decision (decision_1).

Groups combine gender, the number of safe Holt-Laury choices, agreeableness, altruism, and nonconformity. Median splits with randomized ties and a threshold of more than 100 observations yield $K = 13$ groups containing 1,566 records. Released questionnaire measures are carried across gameplay rounds in the author's preparation code.

The mean of the group benchmark HTEs is $\tau_0 = 0.053008$. Their distribution is shown in Figure 3.

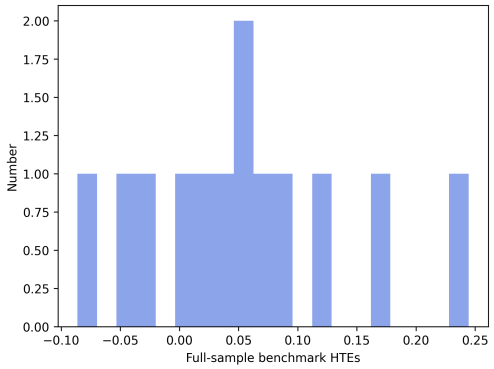
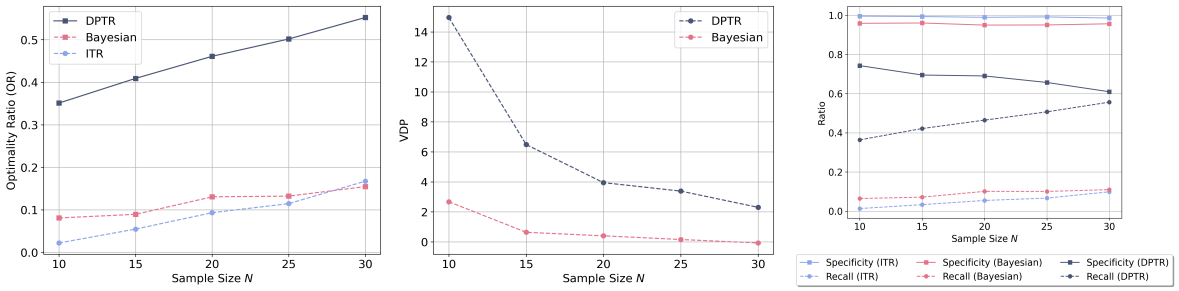


Figure 3 **Histogram of full-sample benchmark HTEs at Banerjee and Mustafi (2026).**

Figure 4 reports mean performance under the common resampling protocol described above.



(a) OR (b) VDP (c) Recall and Specificity

Figure 4 **Performance comparisons at Banerjee and Mustafi (2026) under different sample sizes N.**

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

Risk choices are elicited in period 11 after gameplay; these groups therefore cannot be treated as fully pretreatment targeting strata. Row resampling does not preserve subjects, rematched groups, sessions, or the two collection phases.

3. Social Audience Size as a Reference Point: Evidence from a Field Experiment

We use the 4,000-row public release accompanying Xu et al. (2026) in Management Science. Treatment is the randomized first-phase increase in artificial followers, versus no increase. The outcome is the released first-phase position-opening measure (numpositions1st). A higher value denotes more trading rather than better investment performance.

The two baseline follower strata, at most 9 versus more than 9 followers, define $K = 2$ groups containing all 4,000 released records. We preserve these strata without randomly splitting ties; no other baseline trader covariates are supplied.

The mean of the group benchmark HTEs is $\tau_0 = 4.263559$. Their distribution is shown in Figure 5.

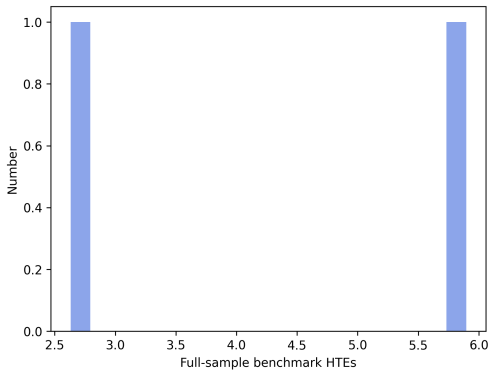
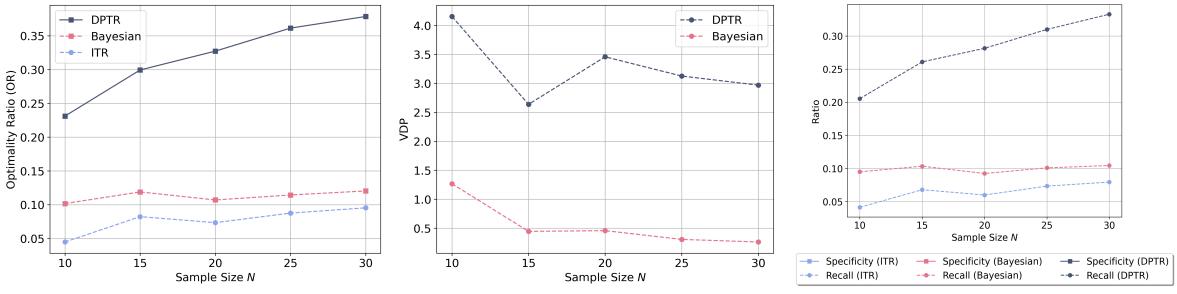


Figure 5 Histogram of full-sample benchmark HTEs at Xu et al. (2026).

Figure 6 reports mean performance under the common resampling protocol described above.



(a) OR (b) VDP (c) Recall and Specificity

Figure 6 Performance comparisons at Xu et al. (2026) under different sample sizes N. Specificity is undefined because all retained benchmark HTEs are positive.

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N. The released position measure includes negative and fractional values, preventing interpretation as literal raw counts. This exercise has only $K = 2$ groups. Both benchmark effects are positive, so specificity is undefined.

4. Performance Ranks, Conformity, and Cooperation: Evidence from a Sweater Factory

We use the public task-level replication data released with Ashraf (2026) in Management Science. We select February-June 2016 jobs in the Private and Public rank arms, leaving 14,961 complete records. Treatment is public rather than private rank feedback. The outcome is log mean knitting time per sweater (ln_meantime).

Groups combine years of schooling, baseline tenure, competitiveness, risk preference, baseline friend count, production month, and the worker-style-size job sequence. Median splits with randomized ties and a threshold of more than 100 observations yield $K = 88$ groups containing 11,623 records. Month and sequence describe task context rather than fixed worker traits.

The mean of the group benchmark HTEs is $\tau_0 = 0.010697$. Their distribution is shown in Figure 7.

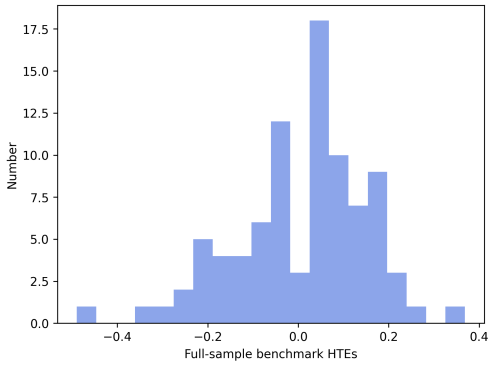


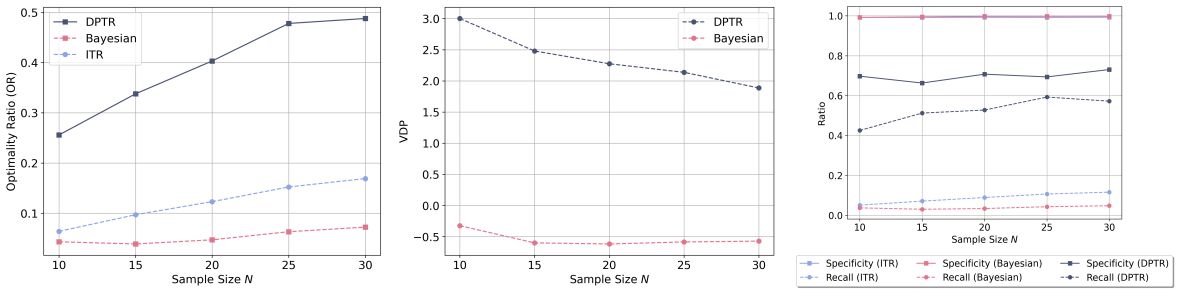
Figure 7 Histogram of full-sample benchmark HTEs at Ashraf (2026).

Figure 8 reports mean performance under the common resampling protocol described above.

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

The ITR and Bayesian specificity curves are close enough to partially overlap at the plotted scale in panel (c).

Higher knitting time means slower production. The reported reward therefore targets increases in recorded time, not productivity improvement. The benchmark effects are unadjusted row differences; block-level dependence is not preserved by resampling.



(a) OR (b) VDP (c) Recall and Specificity

Figure 8 Performance comparisons at Ashraf (2026) under different sample sizes N.

5. Competition Avoidance vs. Herding in Job Search: Evidence from Large-Scale Field Experiments on an Online Job Board

We use the original survey data publicly released with Fradkin et al. (2026) in Management Science, rather than the unavailable Meta platform microdata or the package's simulated platform files. The outcome indicates selection of the vacancy with fewer displayed applications, coding no preference as zero. Treatment is display of AI-estimated job-offer probabilities (is_ai = 1), versus no such probabilities.

After excluding self-reported random responses, we group choice records by age, gender, recent job search, wage employment, question number, whether Brown Co. has fewer applications, and the two displayed applicant-count ranges. Median splits with randomized ties and a threshold of more than 20 observations yield $K = 88$ groups containing 2,986 choice records. Full-sample groupwise mean differences serve as benchmark HTEs.

The mean of the group benchmark HTEs is $\tau_0 = -0.082430$. Their distribution is shown in Figure 9.

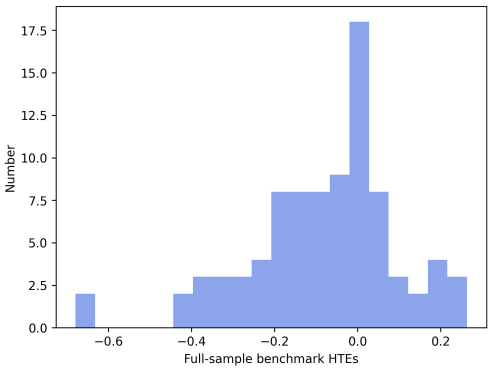
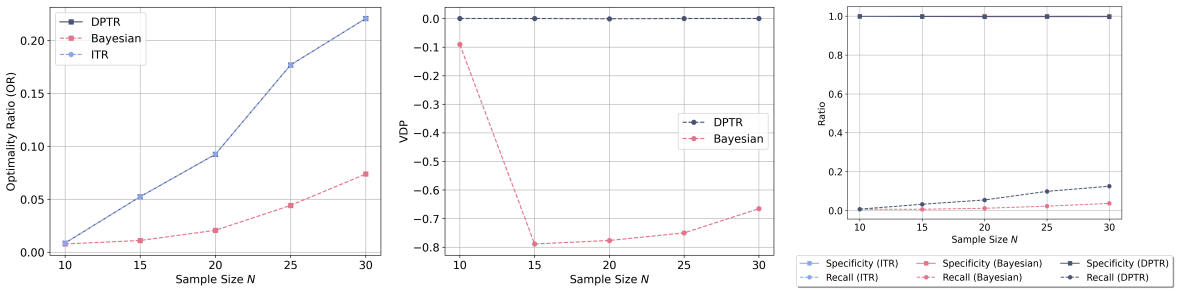


Figure 9 Histogram of full-sample benchmark HTEs at Fradkin et al. (2026).

Figure 10 reports mean performance under the common resampling protocol described above.

Both ITR and DPTR have higher mean OR than the Bayesian benchmark at every tested N. The ITR and DPTR OR curves coincide at $N = 10$, 15, 25, and 30 and nearly overlap at $N = 20$, where



(a) OR (b) VDP (c) Recall and Specificity

Figure 10 Performance comparisons at Fradkin et al. (2026) under different sample sizes N.

their difference is approximately 0.000087. The ITR mean OR is positive at every tested sample size. Their recall curves also nearly overlap, and their specificity curves coincide; the Bayesian specificity curve lies very close to them.

The benchmark anchor is negative. Resampling uses choice records, with repeated choices within respondents, from the survey rather than the paper's platform interventions.

The negligible mean OR gap against ITR (-0.000018, averaged over N) reflects an implementation that usually chooses no pooling. In a diagnostic replay matching the saved five-decimal OR means, the nonnegative pooling scale is zero in 94.6%, 98.7%, 99.9%, 100%, and 100% of resamples as N increases from 10 to 30. At zero scale, DPTR makes the same decisions as ITR, explaining their nearly identical performance despite the negative benchmark anchor. DPTR improves on the Bayesian benchmark at every tested N.

6. Engaging Customers with AI in Online Chats: Evidence from a Randomized Field Experiment

We use the public replication data released with Zhang and Narayandas (2026) in Management Science. The authors describe the release as a random 80% subset with disguised agent identifiers. The outcome is customer sentiment at the end of a chat (end_sentiment). Treatment is access to AI-generated suggestions for the human agent (AI_enabled = 1), versus no access.

Groups use initial customer sentiment, log agent tenure, chatbot comprehension failures, network issues, noncomplaint and repeat-complaint intent indicators, and the first calendar week of AI rollout (January 18-24, 2021). Unclassified intents are coded zero in the two indicators. Median splits with randomized ties and a threshold of more than 100 observations yield $K = 128$ groups containing 205,632 complete conversations. Full-sample subgroup mean differences provide benchmark HTEs.

The mean of the group benchmark HTEs is $\tau_0 = 0.430873$. Their distribution is shown in Figure 11.

Figure 12 reports mean performance under the common resampling protocol described above.

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

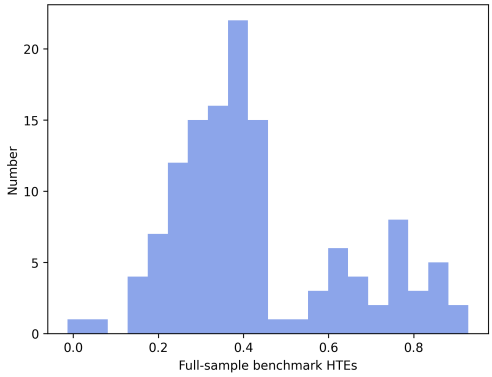
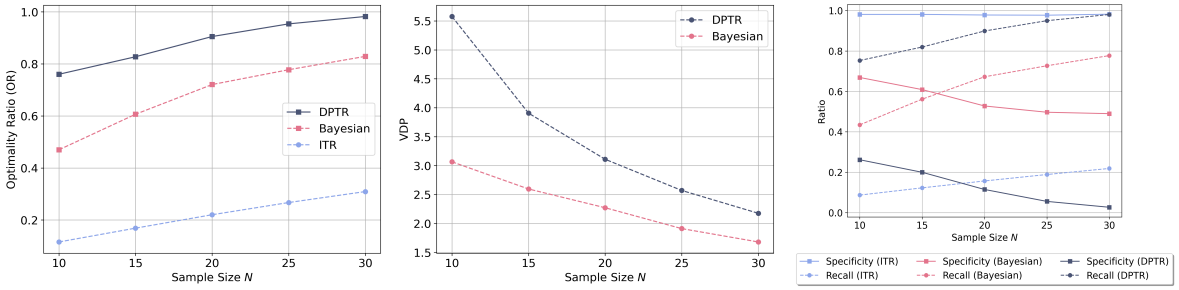


Figure 11 **Histogram of full-sample benchmark HTEs at Zhang and Narayandas (2026).**



(a) OR (b) VDP (c) Recall and Specificity

Figure 12 **Performance comparisons at Zhang and Narayandas (2026) under different sample sizes N.**

AI adoption dates were randomized by agent. The benchmark effects are unadjusted conversation-level differences, and resampling does not preserve agent-level dependence. Intent labels are inferred from conversations, and their pretreatment availability is not established. Specificity uses one non-positive group.

7. The Power of Disagreement: A Field Experiment to Investigate Human-Algorithm Collaboration in Loan Evaluations

We use the public replication data released with Wang et al. (2026) in Management Science. The README identifies a random approximately one-third release, with 7,854 evaluation tasks. We select the rich-information arms, leaving 3,966 tasks. Treatment is disclosure of the algorithm's rationale. The outcome is good_decision, agreement between the final lending decision and actual or imputed repayment performance.

Groups combine loan amount, loan term, interest rate, applicant age, and monthly income level. Median splits with randomized ties and a threshold of more than 100 observations yield $K = 18$ groups containing 2,920 records. These are application characteristics available before evaluation. Full-sample within-group mean differences provide benchmark HTEs.

The mean of the group benchmark HTEs is $\tau_0 = 0.024480$. Their distribution is shown in Figure 13.

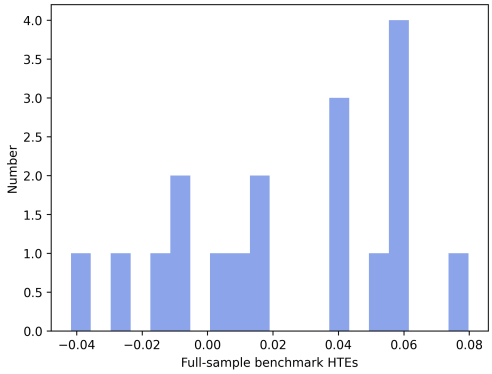
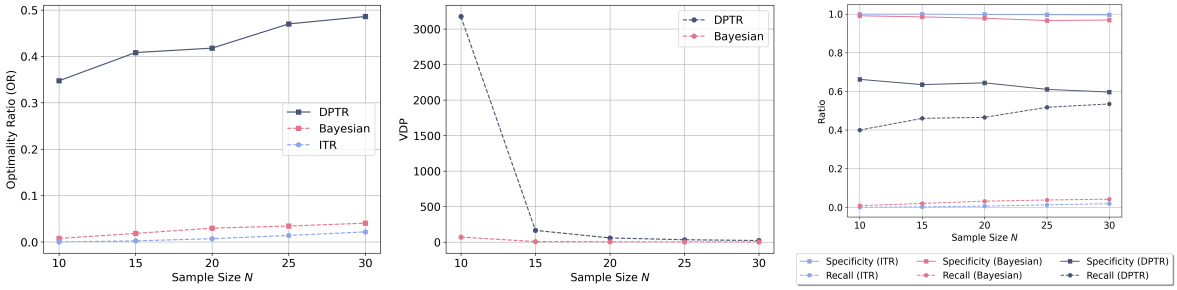


Figure 13 **Histogram of full-sample benchmark HTEs at Wang et al. (2026).**

Figure 14 reports mean performance under the common resampling protocol described above.



(a) OR (b) VDP (c) Recall and Specificity

Figure 14 **Performance comparisons at Wang et al. (2026) under different sample sizes N.**

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

Correctness for denied applicants relies on the authors' imputed counterfactual repayment outcome. Row resampling does not preserve evaluator assignment clusters. The very small ITR reward makes VDP especially large; it is not a loan-return estimate.

8. Human-Algorithm Collaboration with Private Information: Naïve Advice-Weighting Behavior and Mitigation

We use the public Study 2 replication data released with Balakrishnan et al. (2026) in Management Science. Treatment is feature transparency (transparency = 1), versus no transparency (0), excluding training-data transparency (2). The outcome is negative squared final forecast error, $-(\text{task_D} - \text{final})^2$, so that a positive effect represents better prediction accuracy.

We retain participants satisfying the authors' comprehension-check rule. Groups use the high-impact-private-feature indicator, public and private task features, age category, education category, and prior statistics training. Median splits with randomized ties and a threshold of more than 40

observations yield $K = 64$ groups containing 6,860 task records from 343 participants. Full-sample subgroup mean differences serve as benchmark HTEs.

The mean of the group benchmark HTEs is $\tau_0 = 344.749810$. Their distribution is shown in Figure 15.

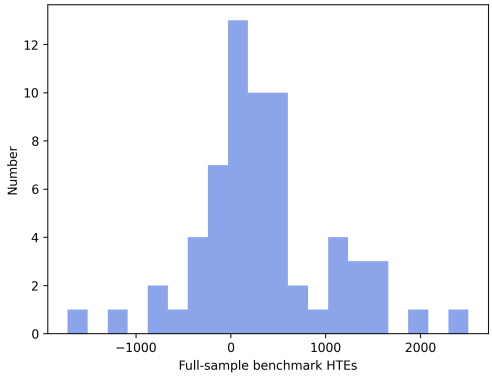
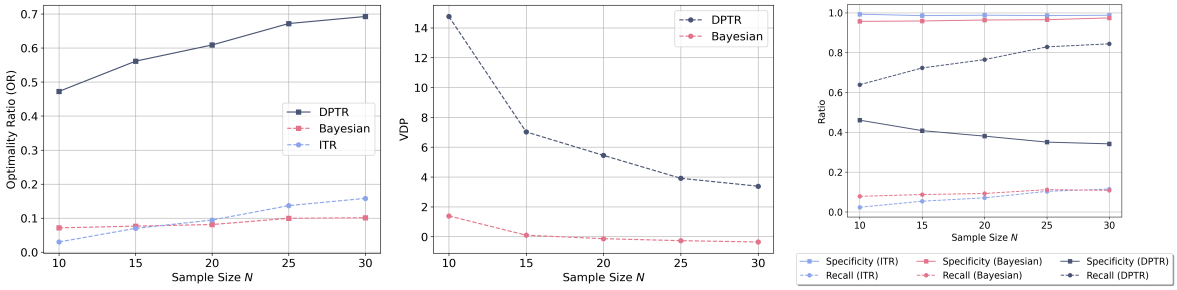


Figure 15 **Histogram of full-sample benchmark HTEs at Balakrishnan et al. (2026).**

Figure 16 reports mean performance under the common resampling protocol described above.



(a) OR (b) VDP (c) Recall and Specificity

Figure 16 **Performance comparisons at Balakrishnan et al. (2026) under different sample sizes N.**

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N. Resampling uses task records, with repeated tasks within participants.

9. Incentives, Framing, and Reliance on Algorithmic Advice: An Experimental Study

We use the public replication data released with Greiner et al. (2026) in Management Science. The outcome is the recorded weight of algorithmic advice, censored to [0, 1]. We compare HumanAI Advice - Fixed Pay with AI Advice - Fixed Pay, retaining 3,473 complete apartment-evaluation records. Both arms use fixed pay; treatment changes the advice framing. Because advice weights are undefined without advice, the AI-advice fixed-pay arm serves as the active control.

Groups combine gender, age, university-degree status, Austrian nationality and prior knowledge of Vienna. Median splits with randomized ties and a threshold of more than 100 observations yield $K = 16$ groups containing 2,423 records from 3,473 complete input records.

The mean of the group benchmark HTEs is $\tau_0 = 0.050636$. Their distribution is shown in Figure 17.

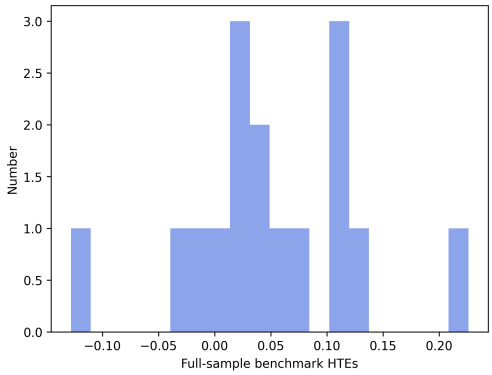
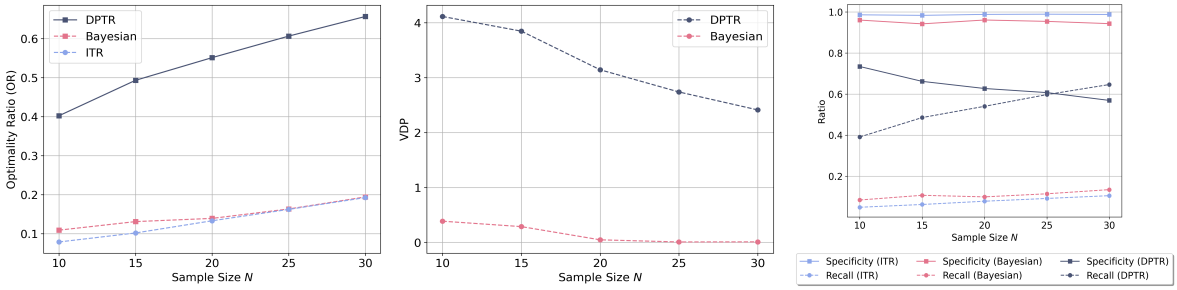


Figure 17 Histogram of full-sample benchmark HTEs at Greiner et al. (2026).

Figure 18 reports mean performance under the common resampling protocol described above.



(a) OR (b) VDP (c) Recall and Specificity

Figure 18 Performance comparisons at Greiner et al. (2026) under different sample sizes N.

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

Repeated evaluations by the same participant are resampled as separate records.

10. Algorithm Reliance: Fast and Slow

We use the public replication data released with Snyder et al. (2026) in Management Science. The outcome is rec_rating, the customer rating of the recommended joke. We compare superior algorithmic advice under an eight-second limit with no algorithmic advice and no eight-second limit, retaining 26,915 complete round-level records. This is a comparison of two complete randomized conditions, so it changes both advice access and the time limit.

Groups combine customer mean joke ratings, customer rating ranges, the participant's STEM-major and male indicators, period, and the participant's pre-existing mean and maximum joke ratings. Median splits with randomized ties and a threshold of more than 100 observations yield $K = 115$ groups containing 25,787 records from 26,915 complete input records.

The mean of the group benchmark HTEs is $\tau_0 = 0.672179$. Their distribution is shown in Figure 19.

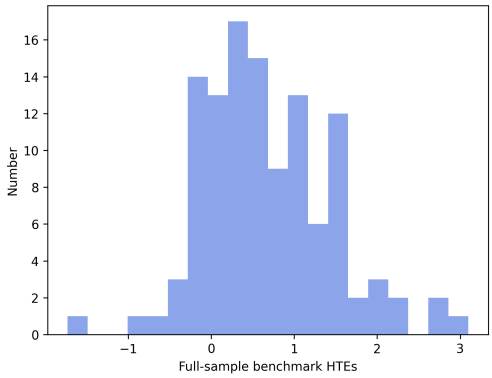
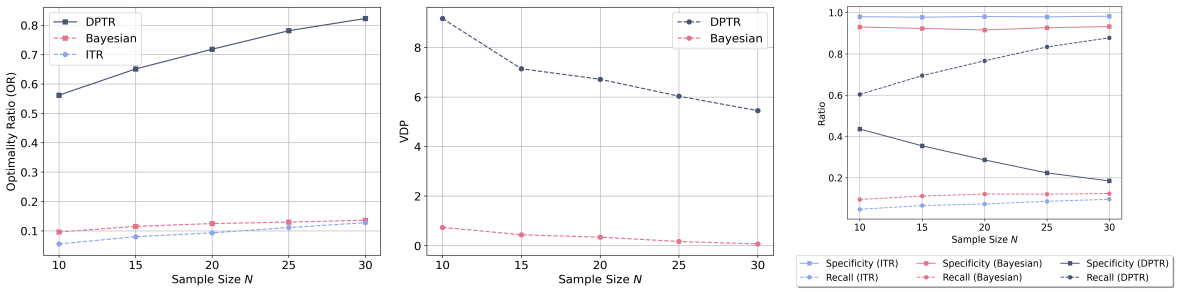


Figure 19 Histogram of full-sample benchmark HTEs at Snyder et al. (2026).

Figure 20 reports mean performance under the common resampling protocol described above.



(a) OR (b) VDP (c) Recall and Specificity

Figure 20 Performance comparisons at Snyder et al. (2026) under different sample sizes N.

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

Repeated participant-round and customer observations are resampled as rows, so participant and customer dependence are not preserved.

11. Strategic Inattention in Product Search

We use the public Study 2 exports released with Hillenbrand and Hippel (2026) in Management Science. We select completed UNIFORM and THEORY sessions, link Prolific metadata, and exclude a participant who completed multiple arms. The outcome is the number of chosen filters (0-5). Treatment is uniform pricing, versus the theoretically optimal discriminatory pricing rule.

Groups combine age, gender, student and employment status, ordinal education, online-shopping frequency, and prior Prolific approvals. Median splits with randomized ties and a threshold of more than 30 observations yield $K = 9$ groups containing 371 records. The complete input comprises 1,440 participant-round records across all 15 rounds; the article principally analyzes the final five.

The mean of the group benchmark HTEs is $\tau_0 = 0.979866$. Their distribution is shown in Figure 21.

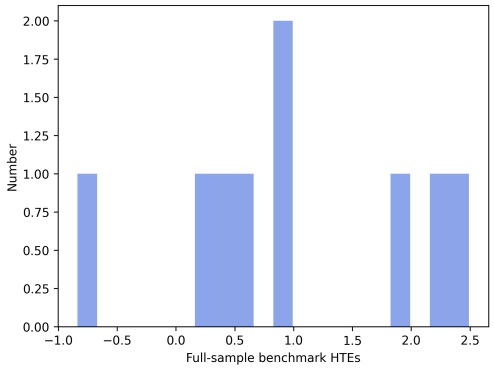
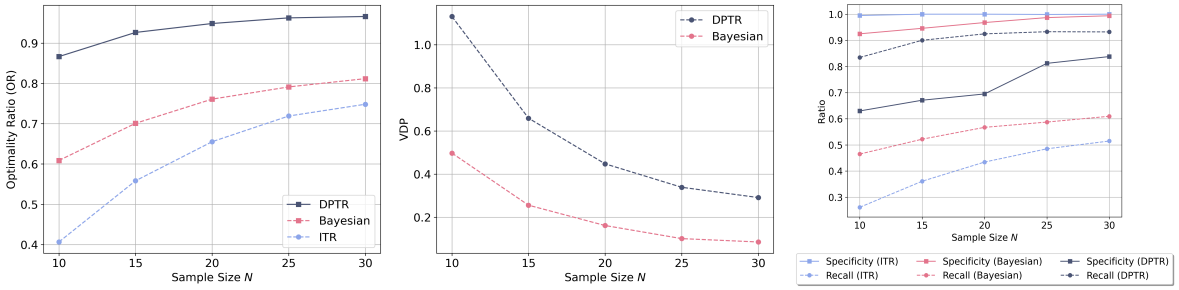


Figure 21 **Histogram of full-sample benchmark HTEs at Hillenbrand and Hippel (2026).**

Figure 22 reports mean performance under the common resampling protocol described above.



(a) OR (b) VDP (c) Recall and Specificity

Figure 22 **Performance comparisons at Hillenbrand and Hippel (2026) under different sample sizes N.**

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

Higher filter use measures search intensity rather than realized consumer surplus. Metadata characteristics and all-round row resampling yield empirical benchmarks, without preserving repeated participants. The retained sample is small relative to the complete input.

12. Signaling Quality: How Refund Bonuses Can Overcome Information Asymmetries in Crowdfunding

We use the public laboratory replication data released with Cason et al. (2025) in Management Science. We select the Random Bonus treatment after the initial no-bonus rounds and aggregate

subject records into 1,440 project-round observations. Treatment is a randomly offered refund bonus. The outcome is successful funding, requiring both investors to invest.

Groups combine true project quality, round, project shape, the investors' negative-report count, mean age, male share, engineering/science share, and mean prior experiment experience. Median splits with randomized ties and a threshold of more than 10 observations yield $K = 39$ groups containing 543 records. Demographics and private signals are aggregated over the two investors.

The mean of the group benchmark HTEs is $\tau_0 = 0.189139$. Their distribution is shown in Figure 23.

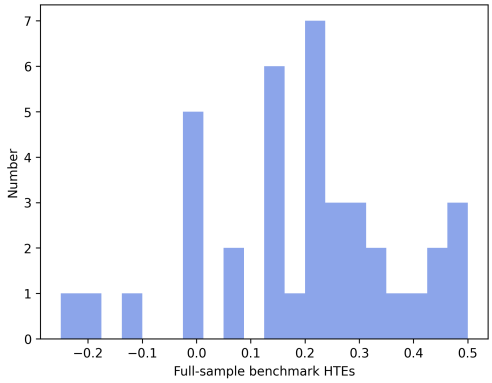
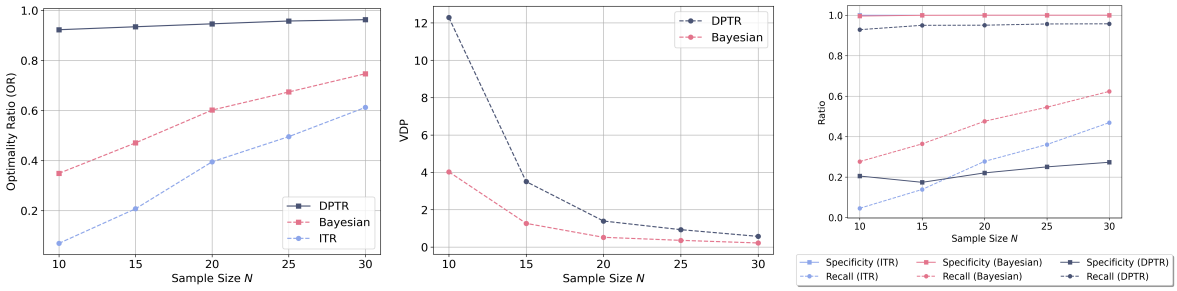


Figure 23 Histogram of full-sample benchmark HTEs at Cason et al. (2025).

Figure 24 reports mean performance under the common resampling protocol described above.



(a) OR (b) VDP (c) Recall and Specificity

Figure 24 Performance comparisons at Cason et al. (2025) under different sample sizes N.

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

The ITR and Bayesian specificity curves coincide for $N \geq 20$ and lie close together at the smaller sample sizes.

These groups mix project context and investor information, including true quality that investors do not fully observe. Project-row resampling does not preserve sessions, matching groups, or repeated investors. Five zero-effect groups enter specificity.

13. Worker Empowerment and Subjective Evaluation: On Building an Effective Conflict Culture

We use the public laboratory replication data released with MacLeod et al. (2025) in Management Science. We select Experiment 4 and retain 2,016 interacting worker-period records. Treatment permits costly conflict after subjective evaluation, versus the same experiment's no-conflict control. The outcome indicates high effort (higheffort).

Groups use period, its square and centered value, the experiment-period product, five-period blocks, and early/middle/late indicators. Median splits with randomized ties and a threshold of more than 180 observations yield $K = 2$ groups containing 755 records from 2,016 complete input records.

The mean of the group benchmark HTEs is $\tau_0 = 0.203964$. Their distribution is shown in Figure 25.

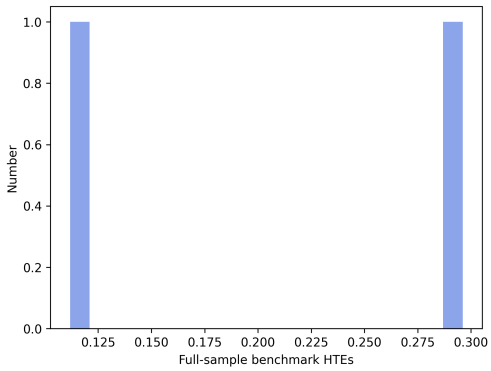
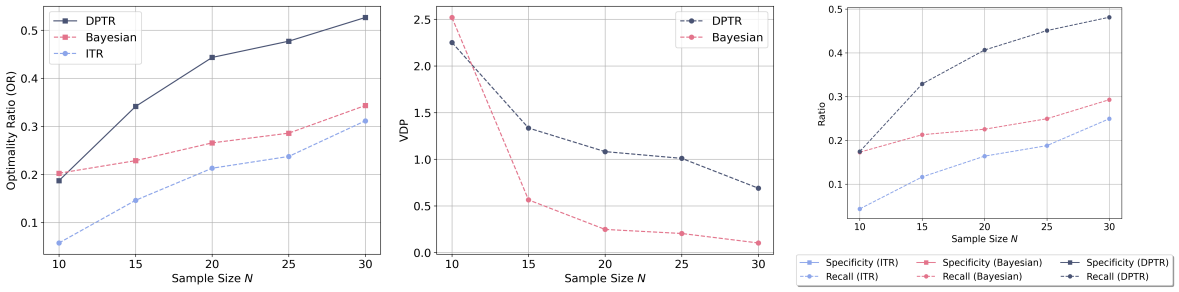


Figure 25 **Histogram of full-sample benchmark HTEs at MacLeod et al. (2025).**

Figure 26 reports mean performance under the common resampling protocol described above.



(a) OR (b) VDP (c) Recall and Specificity

Figure 26 **Performance comparisons at MacLeod et al. (2025) under different sample sizes N. Specificity is undefined because all retained benchmark HTEs are positive.**

DPTR has higher mean OR than ITR at every tested N. The Bayesian benchmark has higher mean OR than DPTR at $N = 10$, whereas DPTR has higher mean OR at $N \geq 15$.

The two benchmark group effects are positive, so specificity is undefined. Conditioning on interaction excludes rejected agreements after assignment. Redundant timing features and randomized ties create artificial partitions; row resampling does not preserve worker or session dependence, and high effort omits conflict costs.

14. Cancel the Deal? An Experimental Study on the Exploitation of Irrational Consumers

We use the public replication data released with Cappelen et al. (2025) in Management Science. The outcome indicates whether a spectator cancels a deal between an informed seller and a misinformed buyer (Cancel). We compare original condition 4, in which the seller both proposed the deal and obfuscated information, with baseline condition 1, which has neither involvement. Reconstructing the outcome from the raw spectator file and treating demographic refusals as missing leaves 1,976 complete records.

Groups combine age, the woman indicator, education category and household-income category. Median splits with randomized ties and a threshold of more than 100 observations yield $K = 8$ groups containing 1,356 records from 1,976 complete input records.

The mean of the group benchmark HTEs is $\tau_0 = 0.043943$. Their distribution is shown in Figure 27.

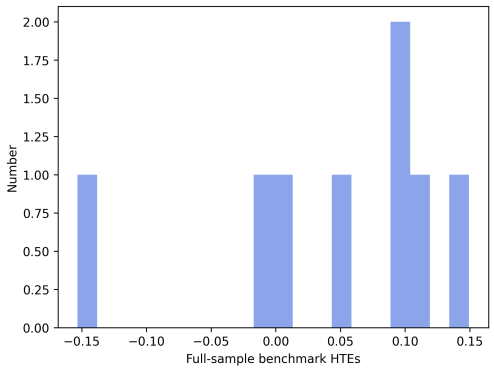


Figure 27 Histogram of full-sample benchmark HTEs at Cappelen et al. (2025).

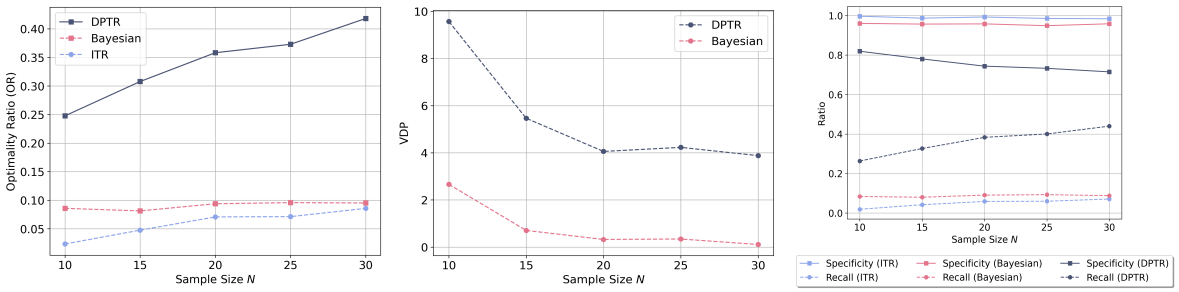
Figure 28 reports mean performance under the common resampling protocol described above.

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

The selected treatment jointly changes the seller-proposal and information-obfuscation dimensions.

15. The Algorithmic Assignment of Incentive Schemes

We use the public replication data released with Opitz et al. (2025) in Management Science, selecting Experiment 2 and excluding the fixed-wage control arm. Treatment is algorithmic assignment to an incentive scheme, versus the on-average best scheme (Bonus Loss). The outcome is the number of correctly completed a/b keypress pairs in ten minutes.



(a) OR (b) VDP (c) Recall and Specificity

Figure 28 Performance comparisons at Cappelen et al. (2025) under different sample sizes N.

Groups combine the 30-second ability test, age, conscientiousness, altruistic donation amount, stated competitiveness, gender, some-college status, and bachelor degree or higher. These measures precede the treatment description. Median splits with randomized ties and a threshold of more than 20 observations yield $K = 135$ groups containing 4,497 records. The complete input contains 6,148 worker records.

The mean of the group benchmark HTEs is $\tau_0 = 45.461092$. Their distribution is shown in Figure 29.

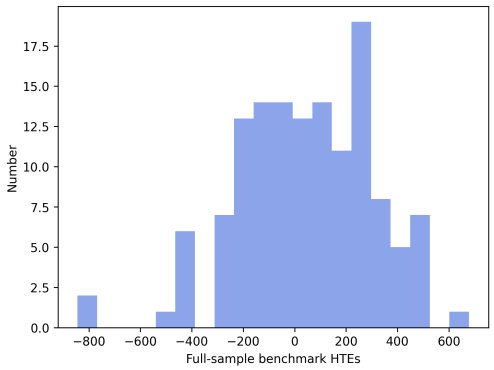


Figure 29 Histogram of full-sample benchmark HTEs at Opitz et al. (2025).

Figure 30 reports mean performance under the common resampling protocol described above.

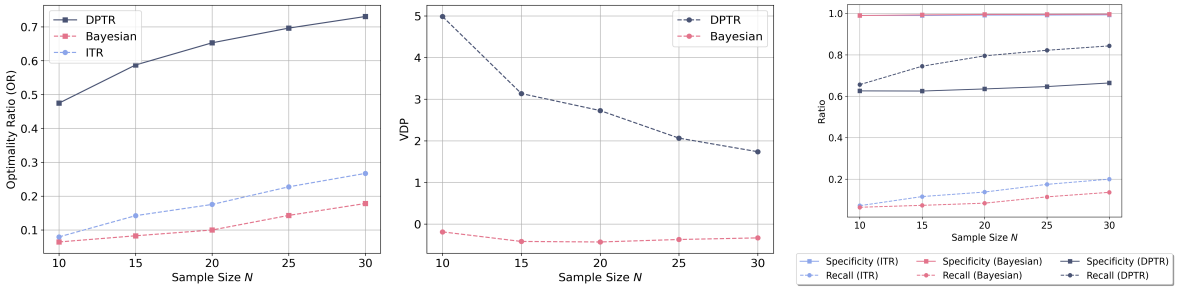
DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

The ITR and Bayesian specificity curves nearly overlap in panel (c).

The benchmark uses unadjusted differences in raw performance. The worker-row bootstrap does not preserve batch-level dependence.

16. The Digital Lives of the Poor: Entertainment Traps and Information Isolation

We use the public replication data released with Ramdas and Sungu (2024) in Management Science. The outcome is attendance at an in-person health camp (camp_attendance). Treatment is assignment



(a) OR (b) VDP (c) Recall and Specificity

Figure 30 Performance comparisons at Opitz et al. (2025) under different sample sizes N.

to a daily-capped data plan, providing 0.5 GB per day, versus a standard four-week plan providing 14 GB without daily caps. The public WhatsApp file contains 3,232 invitation records from 808 participants, linked to the survey by user_id.

Groups use recorded gender, age category, education, income category, self-control, fear of missing out (FOMO), home Wi-Fi access, and phone sharing. Missing FOMO values are filled with the median across merged invitation records. Median splits with randomized ties and a threshold of more than 20 observations yield $K = 23$ groups containing 554 invitation records from 302 participants. Full-sample within-group mean differences provide benchmark HTEs.

The mean of the group benchmark HTEs is $\tau_0 = 0.048170$. Their distribution is shown in Figure 31.

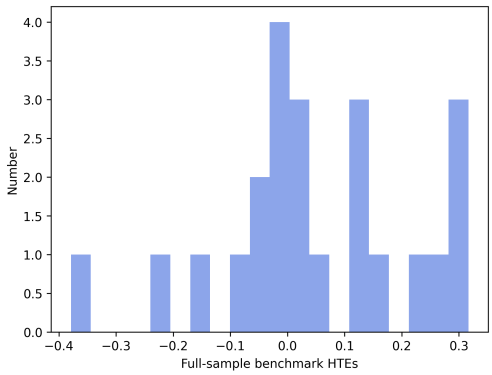
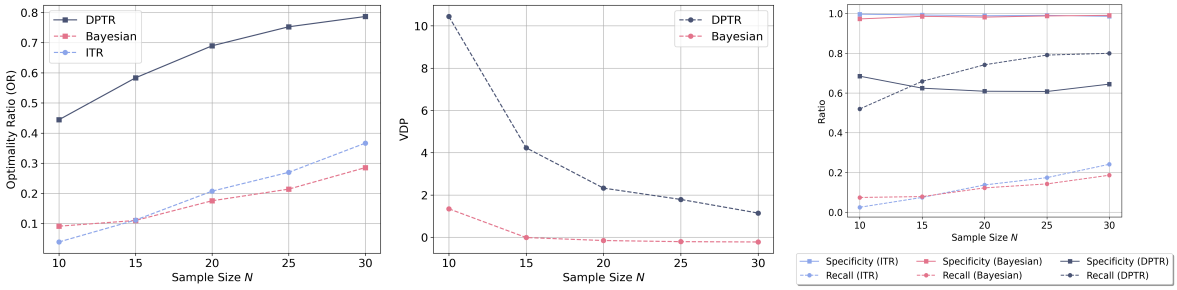


Figure 31 Histogram of full-sample benchmark HTEs at Ramdas and Sungu (2024).

Figure 32 reports mean performance under the common resampling protocol described above. DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

Invitations repeat within participants; record-level resampling does not preserve respondent dependence. The collection timing of the released FOMO score is not established. Specificity includes the two groups with zero benchmark effects.



(a) OR (b) VDP (c) Recall and Specificity

Figure 32 Performance comparisons at Ramdas and Sungu (2024) under different sample sizes N.

17. Participation vs. Effectiveness in Sponsored Tweet Campaigns: A Quality-Quantity Conundrum

We use the public training file excluding task pair 12 released with Peng and Van den Bulte (2024) in Management Science. The outcome indicates whether an endorser retweeted a task. We retain one nonzero compensation condition (payrate = 2) and the zero-compensation control (payrate = 0), leaving 194,658 endorser-task records.

Groups combine log follower count, log verified followers, verified-follower share, log tweet count, log past tasks, prior approval rate, normalized tenure, log referral rewards and log internal-network friends. Nonfinite feature values are replaced by feature medians. Median splits with randomized ties and a threshold of more than 100 observations yield $K = 352$ groups containing 191,957 records from 194,658 complete input records.

The mean of the group benchmark HTEs is $\tau_0 = 0.033326$. Their distribution is shown in Figure 33.

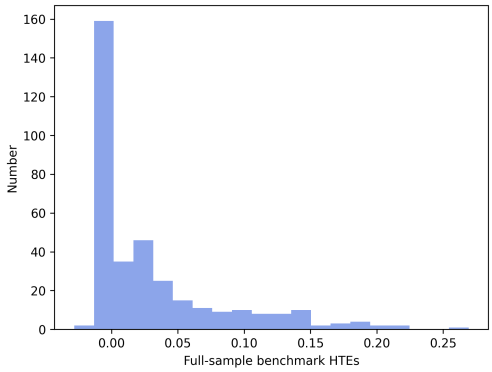
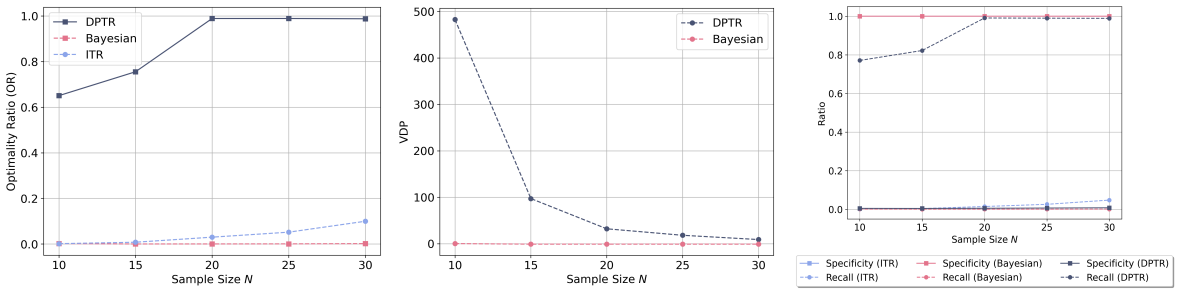


Figure 33 Histogram of full-sample benchmark HTEs at Peng and Van den Bulte (2024).

Figure 34 reports mean performance under the common resampling protocol described above.

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

The ITR and Bayesian specificity curves coincide at $N = 15$, 20, 25, 30 and are distinct at the other tested sample sizes.



(a) OR (b) VDP (c) Recall and Specificity

Figure 34 Performance comparisons at Peng and Van den Bulte (2024) under different sample sizes N.

Repeated endorser-task and task-pair dependence is not preserved by row resampling. Large VDP values arise when the mean ITR reward is small.

18. Discrimination and Economic Expectations

We use the public primary-experiment replication data released with Bazley et al. (2024) in Management Science. The outcome is income uncertainty (earn_sd), the standard deviation of a participant's subjective distribution of earnings growth over the next 12 months. Treatment is the randomized pervasive-discrimination narrative (PervasiveDiscrim = 1), versus the rare-discrimination narrative.

We use the primary experiment only. After treating education refusal as missing and applying complete-case selection, 1,421 of the 1,529 released participants remain. Groups combine age category, the woman indicator, education, income category, marital status, and the White-race indicator. Median splits with randomized ties and a threshold of more than 20 observations yield $K = 30$ groups containing 909 participants.

The mean of the group benchmark HTEs is $\tau_0 = 0.500357$. Their distribution is shown in Figure 35.

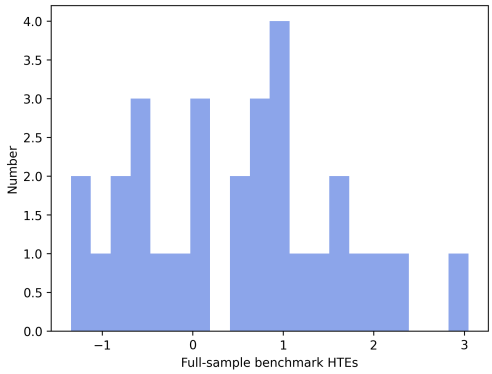
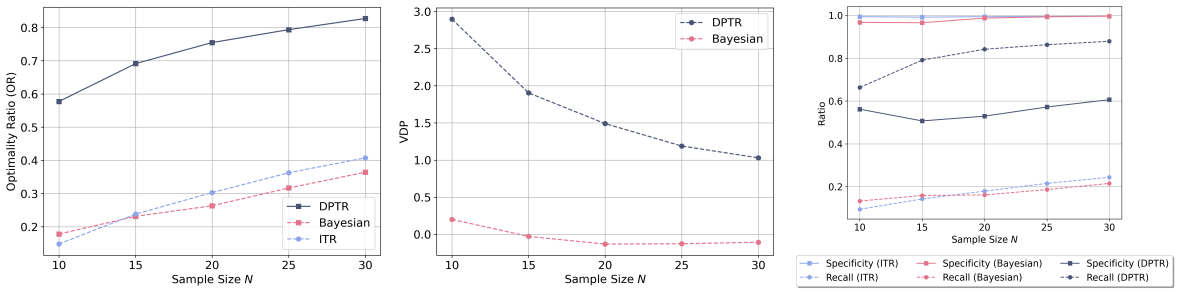


Figure 35 Histogram of full-sample benchmark HTEs at Bazley et al. (2024).

Figure 36 reports mean performance under the common resampling protocol described above. DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.



(a) OR (b) VDP (c) Recall and Specificity

Figure 36 Performance comparisons at Bazley et al. (2024) under different sample sizes N.

The ITR and Bayesian specificity curves lie close together in panel (c) and partially overlap at the plotted scale.

Higher reward here represents a greater increase in subjective income uncertainty, rather than an improvement in welfare. Demographic nonresponse and the release's completion-time restrictions limit the interpretation of these complete-case subgroup benchmarks.

19. Motivating Experts to Contribute to Digital Public Goods: A Personalized Field Experiment on Wikipedia

We use the public replication data released with Chen et al. (2024b) in Management Science, selecting expert-article records whose initial invitation email was opened. The outcome indicates an actual contribution to an article. We compare HighView-Cite (original arm 5) with AvgView-NoCite (arm 1), retaining 6,399 complete records. Treatment jointly changes the readership message and the citation incentive. Both conditions recommend highly viewed articles.

Groups combine expert-article cosine similarity, log abstract views, log article length, English-speaking affiliation, top-10% expert rank, research overlap, and article quality and importance classes. Median splits with randomized ties and a threshold of more than 100 observations yield $K = 2$ groups containing 215 records from 6,399 complete input records.

The mean of the group benchmark HTEs is $\tau_0 = -0.028770$. Their distribution is shown in Figure 37.

Figure 38 reports mean performance under the common resampling protocol described above.

DPTR has lower mean OR than both ITR and the Bayesian benchmark at every tested N.

The ITR and Bayesian specificity curves coincide at every tested N.

The analysis has two benchmark groups. Opening the email is observed after assignment, and row resampling does not preserve expert/article dependence. VDP is undefined at $N = 10$ and $N = 15$ because the mean ITR reward is zero; the reported DPTR mean OR is negative at every tested N.

With $K = 2$, one positive and one negative benchmark HTE provide little information for estimating the negative benchmark anchor. DPTR selects the negative-effect group in 17.1%–28.3% of

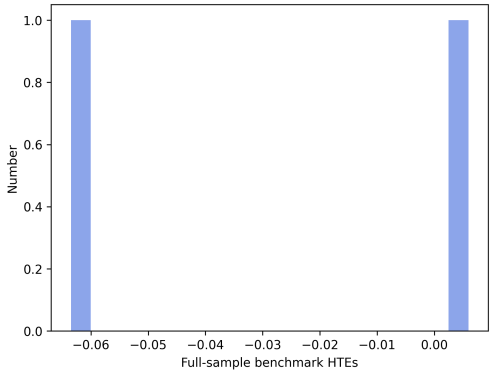
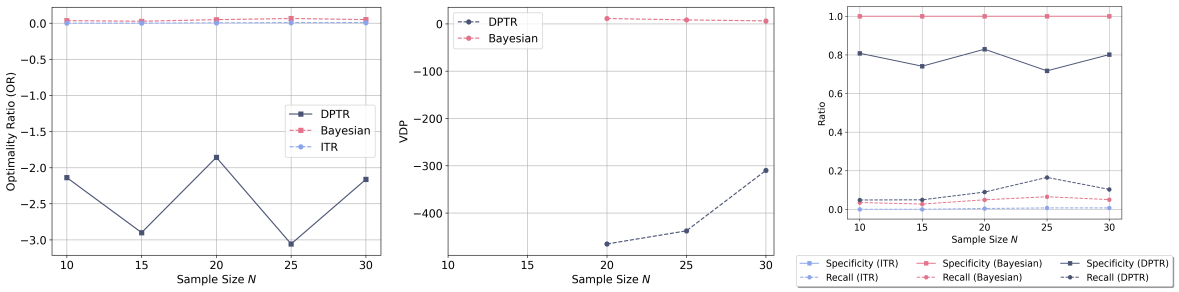


Figure 37 **Histogram of full-sample benchmark HTEs at Chen et al. (2024b).**



(a) OR (b) VDP (c) Recall and Specificity

Figure 38 **Performance comparisons at Chen et al. (2024b) under different sample sizes N. VDP is undefined at $N = 10, 15$ because the mean ITR reward is zero.**

resamples across the tested N, whereas neither comparator selects it in any of the 1,000 resamples per setting. For a finite nonnegative scale, the implemented DPTR criterion is a convex combination of the same Welch lower bound used by ITR and the estimated anchor. Since ITR never selects this group, its lower bound is always nonpositive. A DPTR selection therefore requires a positive estimated anchor. The saved selections thus establish anchor sign switching. The group's size-weighted benchmark effect is -0.0327796, against an empirical oracle reward of only 0.00287929. Selecting this group reduces benchmark reward, and the small oracle magnifies the negative OR.

20. Automation Enables Specialization: Field Evidence

We use the public replication data released with Gong and Png (2024) in Management Science. The released cashier data disguise identifiers and replace basket size and value with interval averages. The outcome is log scanning speed (itempm_tln). Treatment is scan-only checkout (scan = 1), where customers pay at a self-pay kiosk, versus conventional checkout.

We partition transactions using transformed basket value, basket size and average item value, the Wednesday indicator, the proportion of closed counters, transformed time on shift, recent transaction volume, hour, and three store indicators. Median splits with randomized ties retain groups containing

more than 200 observations. From 152,246 complete records, we obtain $K = 222$ groups containing 63,557 transactions. Groupwise full-sample mean differences provide the HTE benchmarks.

The mean of the group benchmark HTEs is $\tau_0 = 0.085601$. Their distribution is shown in Figure 39.

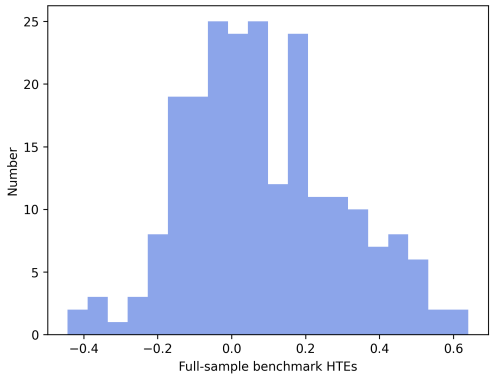
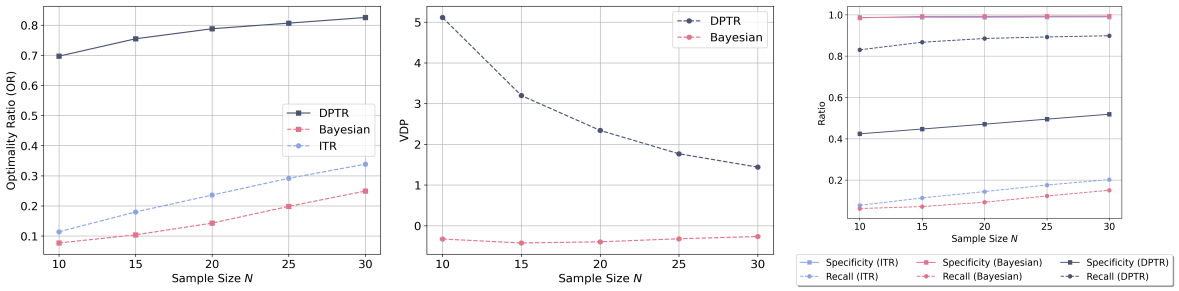


Figure 39 Histogram of full-sample benchmark HTEs at Gong and Png (2024).

Figure 40 reports mean performance under the common resampling protocol described above.



(a) OR (b) VDP (c) Recall and Specificity

Figure 40 Performance comparisons at Gong and Png (2024) under different sample sizes N.

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

The ITR and Bayesian specificity curves nearly overlap in panel (c); their differences are small at the plotted scale.

The benchmark uses unadjusted subgroup mean differences. Transaction-level resampling does not preserve cashier dependence.

21. Intrachoice Dynamics Shape Social Decisions

We use the public Study 2 replication data released with Chen et al. (2024a) in Management Science, drawn from the keyboard-response experiment of Chen and Krajbich (2018). The outcome is prosocial

choice, coded as 1 minus SelfishDecision. Treatment is the time-pressure condition, versus the time-free condition. We exclude delay trials and retain GameID ≤ 50, giving 10,200 decisions from 102 participants.

Groups use absolute differences in own and others' payoffs, maximum and minimum own and others' payoffs, the male indicator, and total payoffs for options A and B. Median splits with randomized ties and a threshold of more than 100 observations yield $K = 32$ groups containing 6,160 decision records. Full-sample within-group mean differences provide benchmark HTEs.

The mean of the group benchmark HTEs is $\tau_0 = -0.095483$. Their distribution is shown in Figure 41.

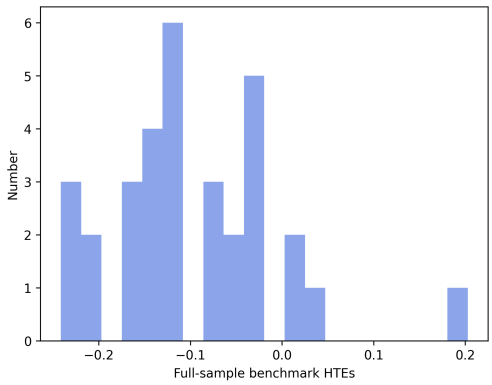
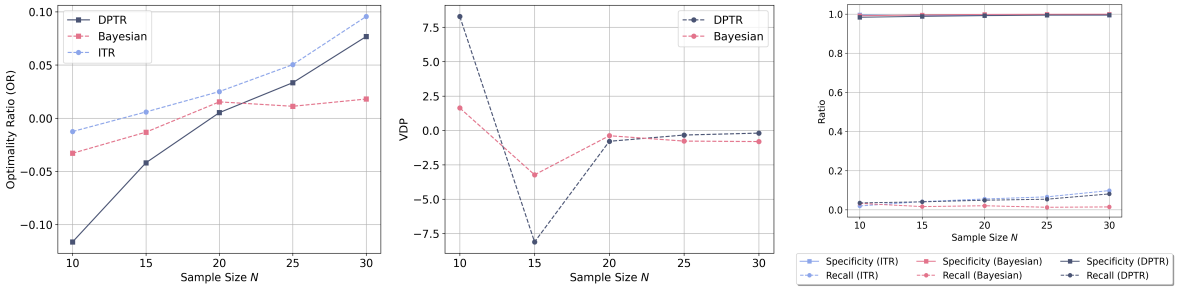


Figure 41 Histogram of full-sample benchmark HTEs at Chen et al. (2024a).

Figure 42 reports mean performance under the common resampling protocol described above.



(a) OR (b) VDP (c) Recall and Specificity

Figure 42 Performance comparisons at Chen et al. (2024a) under different sample sizes N.

DPTR has lower mean OR than ITR at every tested N.

The three specificity curves lie close together in panel (c) and partially overlap at the plotted scale.

Conditions are repeated within participants, and matched game IDs have one-unit shifts in payoff levels across conditions. At $N = 10$, ITR's mean reward is negative, so a positive VDP ratio does not indicate a reward gain.

Only four of the 32 groups have positive benchmark HTEs, and the benchmark anchor is -0.095483. In a diagnostic replay matching the saved five-decimal OR means, the noise-corrected heterogeneity estimate is nonpositive in 44.7% of resamples at $N = 10$ and 20.7% at $N = 30$. The mean OR gap against ITR, averaged over the five N values, is -0.041442. These diagnostics show substantial finite-sample uncertainty in the pooling inputs for this predominantly negative-effect portfolio.

22. Charitable Giving by the Poor: A Field Experiment in Kyrgyzstan

We use the public replication data released with Adena et al. (2024) in Management Science. The release omits firm-private controls. The outcome is positive-donation participation (donation_dummy). We compare one-to-one donation matching combined with a local-benefits message with an unconditional lead gift and no local-benefits message. These are two complete cells of the original two-by-two experiment; 77,666 complete conservative-sample client records remain.

Groups use loan cycle, age, gender, the recorded vocational/student education dummy, agriculture/trade/services occupation indicators, the single-client marital-status dummy, urban location, project proximity and office client counts. Median splits with randomized ties and a threshold of more than 100 observations yield $K = 54$ groups containing 6,489 records from 77,666 complete input records.

The mean of the group benchmark HTEs is $\tau_0 = 0.023325$. Their distribution is shown in Figure 43.

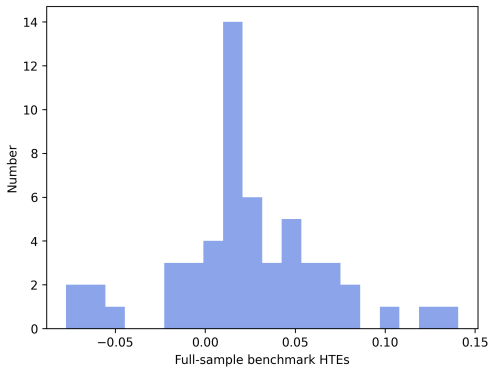
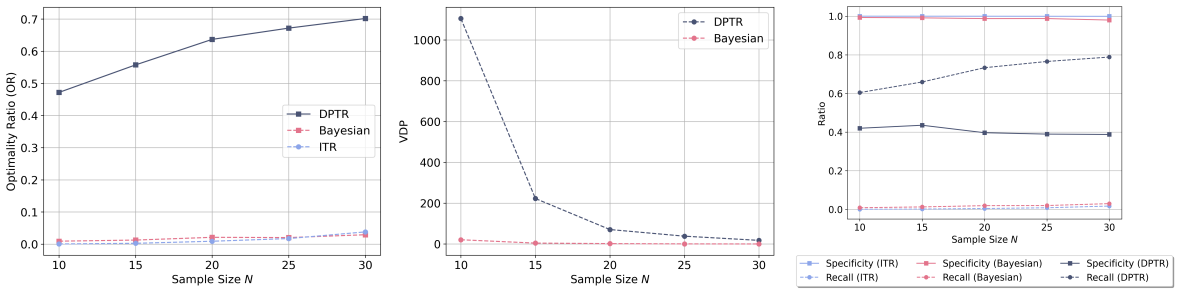


Figure 43 Histogram of full-sample benchmark HTEs at Adena et al. (2024).

Figure 44 reports mean performance under the common resampling protocol described above.

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

Treatments were assigned at office level; the client-row bootstrap does not preserve office-level dependence.



(a) OR (b) VDP (c) Recall and Specificity

Figure 44 Performance comparisons at Adena et al. (2024) under different sample sizes N.

23. The Effects of Online Social Identity Signals on Retailer Demand

We use the public Study 1 replication data released with Babar et al. (2023) in Management Science, rather than the licensed archival foot-traffic data. We compare the woman-owned business label alone (arm W) with neither ownership label (arm C), retaining 8,064 complete profile evaluations. Neither selected arm displays a Black-owned label. The outcome is stated willingness to visit (Self, 1-5).

Groups combine respondent race, gender, political orientation, age, income, latitude, longitude and a liberal indicator (politics ≤ 1). Coordinates are respondent locations. Median splits with randomized ties and a threshold of more than 30 observations yield $K = 107$ groups containing 5,396 records from 8,064 complete input records.

The mean of the group benchmark HTEs is $\tau_0 = -0.000788$. Their distribution is shown in Figure 45.

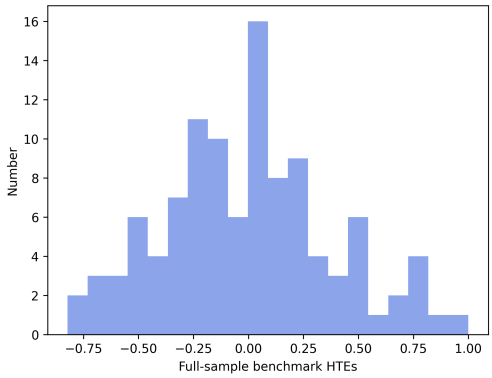


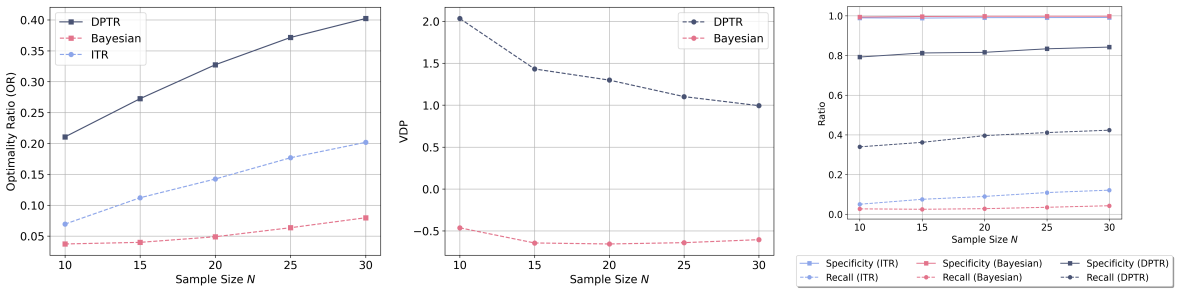
Figure 45 Histogram of full-sample benchmark HTEs at Babar et al. (2023).

Figure 46 reports mean performance under the common resampling protocol described above.

DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

The ITR and Bayesian specificity curves nearly overlap at the plotted scale.

The willingness outcome is stated intention rather than an observed store visit. Repeated profiles from each respondent are resampled as rows, so respondent dependence is not preserved.



(a) OR (b) VDP (c) Recall and Specificity

Figure 46 Performance comparisons at Babar et al. (2023) under different sample sizes N.

24. Putting Teams into the Gig Economy: A Field Experiment at a Ride-Sharing Platform

We use the public replication package accompanying Ai et al. (2023) in Management Science. The authors explicitly state that all released data are synthetic; they are not the confidential field observations reported in the article. We select 2,100 teamed or control drivers. Treatment is team formation plus interteam contests, versus the contacted control group. The outcome is the change in daily revenue during contest days relative to the precontest period.

Groups combine age, platform tenure, locality, gender, prior revenue, and precontest mean, maximum, and minimum trip counts, computed from panel days before day 15. Median splits with randomized ties and a threshold of more than 25 observations yield $K = 4$ groups containing 112 records.

The mean of the group benchmark HTEs is $\tau_0 = 5.806108$. Their distribution is shown in Figure 47.

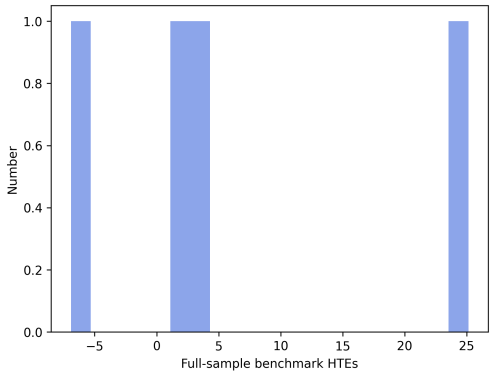
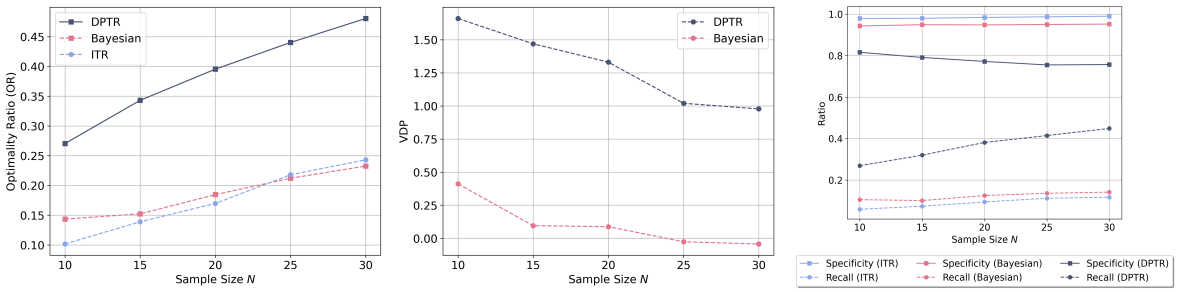


Figure 47 Histogram of full-sample benchmark HTEs at Ai et al. (2023).

Figure 48 reports mean performance under the common resampling protocol described above. DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.

The retained groups contain 112 synthetic drivers, resampled individually despite the team-based treatment.



(a) OR (b) VDP (c) Recall and Specificity

Figure 48 Performance comparisons at Ai et al. (2023) under different sample sizes N.

25. Design and Evaluation of Optimal Free Trials

We use the public replication data released with Yoganarasimhan et al. (2023) in Management Science. The release documentation and dataRandomizer.py state that non-ID column values are replaced by random category codes. These data do not preserve the original experimental joint distribution. We use subscribed as the outcome and compare released trial_length code 0 (treatment) with code 2 (control), dropping code 1. These codes cannot be identified as actual 7- versus 30-day trials.

Groups are formed using code-zero indicators for country, business segment, sign-up category, sign-up application, skill, job and operating system, together with arch64. Median splits with randomized ties and a threshold of more than 100 observations produce $K = 256$ groups containing 225,177 records. Their full-sample mean differences are numerical benchmarks for this released file, not the original trial's causal HTEs.

The mean of the group benchmark HTEs is $\tau_0 = -0.001528$. Their distribution is shown in Figure 49.

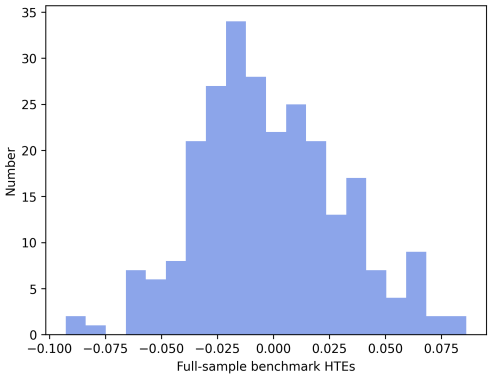
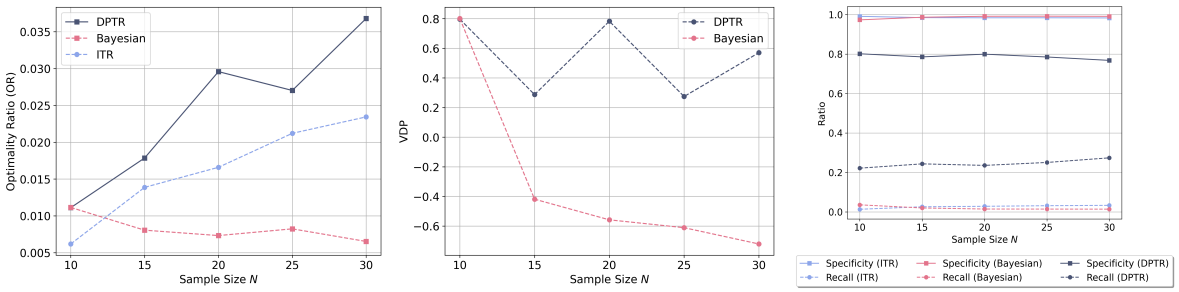


Figure 49 Histogram of full-sample benchmark HTEs at Yoganarasimhan et al. (2023).

Figure 50 reports mean performance under the common resampling protocol described above.



(a) OR (b) VDP (c) Recall and Specificity

Figure 50 Performance comparisons at Yoganarasimhan et al. (2023) under different sample sizes N.

DPTR has higher mean OR than ITR at every tested N. The Bayesian benchmark has slightly higher mean OR than DPTR at $N = 10$, whereas DPTR has higher mean OR at $N \geq 15$.

26. The Worst-First Heuristic: How Decision Makers Manage Conjunctive Risk

We use the public cleaned Study 1 replication data released with Lewis et al. (2023) in Management Science. The 252 participants make eight decisions each. Treatment presents an opportunity to improve the weaker rather than stronger requirement of a conjunctive lottery. The outcome indicates willingness to perform the required typing effort.

Groups combine required typing, bonus amount, baseline stronger/weaker probability ratio, display order, age clipped at 100, and gender. Median splits with randomized ties and a threshold of more than 30 observations yield $K = 30$ groups containing 1,387 records.

The mean of the group benchmark HTEs is $\tau_0 = 0.104270$. Their distribution is shown in Figure 51.

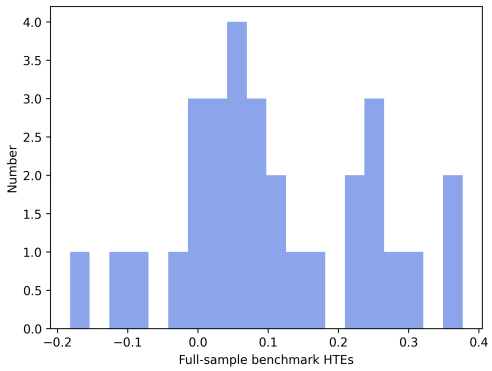
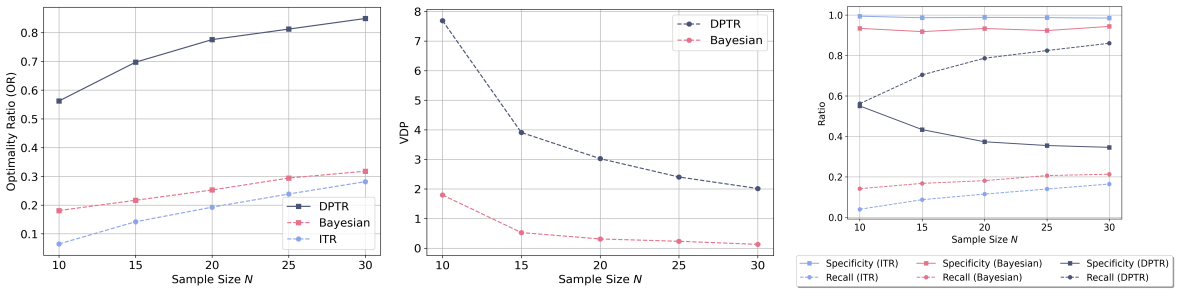


Figure 51 Histogram of full-sample benchmark HTEs at Lewis et al. (2023).

Figure 52 reports mean performance under the common resampling protocol described above. DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.



(a) OR (b) VDP (c) Recall and Specificity

Figure 52 Performance comparisons at Lewis et al. (2023) under different sample sizes N.

Demographics were collected after the decisions. More accepted typing opportunities measure the framing response rather than greater expected welfare. Row resampling does not preserve paired lotteries or repeated participants; one zero-effect group enters specificity.

27. Explaining Excess Entry in Winner-Take-All Markets

We use the public replication data released with Laferrière et al. (2023) in Management Science. The outcome is market entry (entry). Treatment is the winner-take-all market (wta = 1), versus the expected-payoff-equivalent market-entry game with the prize divided among entrants.

We select the main experiment and its first 20-period sequence, retaining 3,080 participant-period decisions from 154 participants in seven sessions. Period is the sole grouping feature. Its median split yields early periods 1-10 and late periods 11-20, producing $K = 2$ groups, each with 1,540 records and more than 100 observations.

The mean of the group benchmark HTEs is $\tau_0 = 0.143333$. Their distribution is shown in Figure 53.

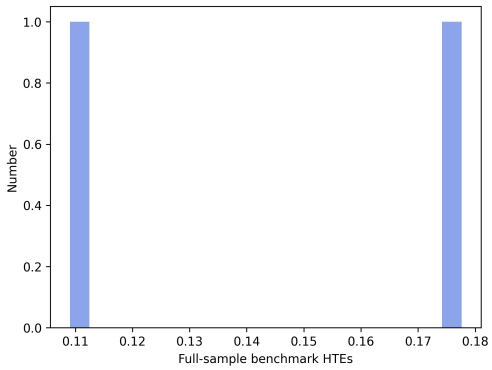
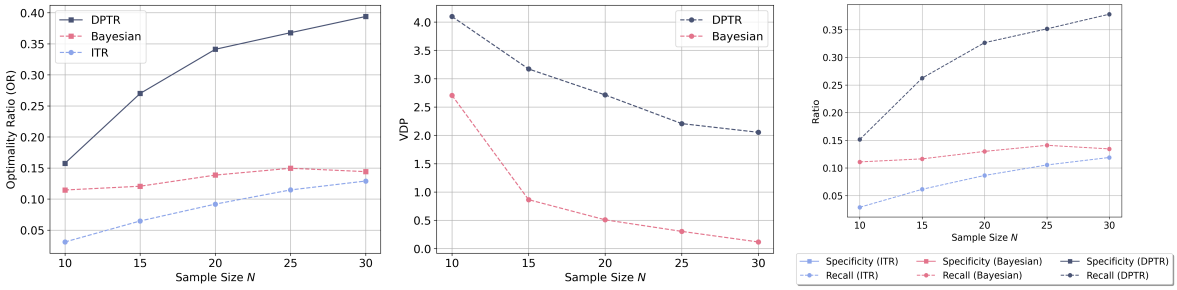


Figure 53 Histogram of full-sample benchmark HTEs at Laferrière et al. (2023).

Figure 54 reports mean performance under the common resampling protocol described above. DPTR achieves higher mean OR than ITR and the Bayesian benchmark at every tested N.



(a) OR (b) VDP (c) Recall and Specificity

Figure 54 Performance comparisons at Laferrière et al. (2023) under different sample sizes N. Specificity is undefined because all retained benchmark HTEs are positive.

This small-K exercise resamples participant decisions as rows, omitting repeated-participant, strategic-interaction, and shared market/session dependence.

References

Adena M, Hakimov R, Huck S (2024) Charitable giving by the poor: A field experiment in kyrgyzstan. Management Science 70(1):633–646, URL http://dx.doi.org/10.1287/mnsc.2023.4702.

Ai W, Chen Y, Mei Q, Ye J, Zhang L (2023) Putting teams into the gig economy: A field experiment at a ride-sharing platform. Management Science 69(9):5336–5353, URL http://dx.doi.org/10.1287/mnsc.2022.4624.

Ashraf A (2026) Performance ranks, conformity, and cooperation: Evidence from a sweater factory. Management Science 72(4):2955–2975, URL http://dx.doi.org/10.1287/mnsc.2023.01381.

Babar Y, Adeli AM, Burtch G (2023) The effects of online social identity signals on retailer demand. Management Science 69(12):7335–7346, URL http://dx.doi.org/10.1287/mnsc.2023.01168.

Balakrishnan M, Ferreira KJ, Tong J (2026) Human-algorithm collaboration with private information: Naïve advice-weighting behavior and mitigation. Management Science 72(1):265–284, URL http://dx.doi.org/10.1287/mnsc.2022.03850.

Banerjee R, Mustafi P (2026) Using social recognition to address the gender difference in volunteering for low-promotability tasks. Management Science 72(5):3997–4011, URL http://dx.doi.org/10.1287/mnsc.2022.04017.

Bazley W, Bonaparte Y, Korniotis G, Kumar A (2024) Discrimination and economic expectations. Management Science 70(6):4115–4131, URL http://dx.doi.org/10.1287/mnsc.2022.02276.

Cappelen AW, Meissner S, Tungodden B (2025) Cancel the deal? an experimental study on the exploitation of irrational consumers. Management Science 71(5):3983–3995, URL http://dx.doi.org/10.1287/mnsc.2023.01050.

Cason TN, Tabarrok A, Zubrickas R (2025) Signaling quality: How refund bonuses can overcome information asymmetries in crowdfunding. Management Science 71(7):5933–5947, URL http://dx.doi.org/10.1287/mnsc.2023.03923.

Chen F, Zhu Z, Shen Q, Krajbich I, Hare TA (2024a) Intrachoice dynamics shape social decisions. Management Science 70(2):1137–1153, URL http://dx.doi.org/10.1287/mnsc.2023.4732.

Chen Y, Farzan R, Kraut R, YeckehZaare I, Zhang AF (2024b) Motivating experts to contribute to digital public goods: A personalized field experiment on wikipedia. Management Science 70(5):3264–3280, URL http://dx.doi.org/10.1287/mnsc.2023.4852.

Fradkin A, Bhole M, Horton JJ (2026) Competition avoidance vs. herding in job search: Evidence from large-scale field experiments on an online job board. Management Science 72(2):1305–1323, URL http://dx.doi.org/10.1287/mnsc.2023.02483.

Gong J, Png IPL (2024) Automation enables specialization: Field evidence. Management Science 70(3):1580–1595, URL http://dx.doi.org/10.1287/mnsc.2023.4760.

Greiner B, Grünwald P, Lindner T, Lintner G, Wiernsperger M (2026) Incentives, framing, and reliance on algorithmic advice: An experimental study. Management Science 72(1):302–322, URL http://dx.doi.org/10.1287/mnsc.2022.02777.

Hillenbrand A, Hippel S (2026) Strategic inattention in product search. Management Science 72(1):693–722, URL http://dx.doi.org/10.1287/mnsc.2022.03413.

Laferrière V, Staubli D, Thöni C (2023) Explaining excess entry in winner-take-all markets. Management Science 69(2):1050–1069, URL http://dx.doi.org/10.1287/mnsc.2022.4397.

Lewis J, Feiler D, Adner R (2023) The worst-first heuristic: How decision makers manage conjunctive risk. Management Science 69(3):1575–1596, URL http://dx.doi.org/10.1287/mnsc.2022.4411.

MacLeod WB, Valle Lara V, Zehnder C (2025) Worker empowerment and subjective evaluation: On building an effective conflict culture. Management Science 71(6):4643–4668, URL http://dx.doi.org/10.1287/mnsc.2022.03085.

Opitz S, Sliwka D, Vogelsang T, Zimmermann T (2025) The algorithmic assignment of incentive schemes. Management Science 71(2):1546–1563, URL http://dx.doi.org/10.1287/mnsc.2022.03362.

Peng J, Van den Bulte C (2024) Participation vs. effectiveness in sponsored tweet campaigns: A quality-quantity conundrum. Management Science 70(11):7961–7983, URL http://dx.doi.org/10.1287/mnsc.2019.01897.

Ramdas K, Sungu A (2024) The digital lives of the poor: Entertainment traps and information isolation. Management Science 70(12):9089–9100, URL http://dx.doi.org/10.1287/mnsc.2024.4989.

Rogers P (2026) Retargeting social media outreach for SNAP take-up. Management Science 72(6):5381–5387, URL http://dx.doi.org/10.1287/mnsc.2024.06312.

Snyder C, Keppler S, Leider S (2026) Algorithm reliance: Fast and slow. Management Science 72(1):368–385, URL http://dx.doi.org/10.1287/mnsc.2023.01989.

Wang H, Zhang Y, Lu T (2026) The power of disagreement: A field experiment to investigate human-algorithm collaboration in loan evaluations. Management Science 72(1):96–118, URL http://dx.doi.org/10.1287/mnsc.2022.03844.

Xu X, Wang Q, Liu Y, Qiu L (2026) Social audience size as a reference point: Evidence from a field experiment. Management Science 72(5):4298–4318, URL http://dx.doi.org/10.1287/mnsc.2022.03144.

Yoganarasimhan H, Barzegary E, Pani A (2023) Design and evaluation of optimal free trials. Management Science 69(6):3220–3240, URL http://dx.doi.org/10.1287/mnsc.2022.4507.

Zhang S, Narayandas D (2026) Engaging customers with AI in online chats: Evidence from a randomized field experiment. Management Science 72(1):73–95, URL http://dx.doi.org/10.1287/mnsc.2022.03920.